

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN DRAINAGE SYSTEM

MODULE - 4

**PENINSULAR RIVER
SYSTEM**

THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

There are a large number of river systems in the Peninsular drainage. A brief account of the major Peninsular river systems is given below:

POINT NO.1

THE MAHANADI

- 1 The Mahanadi rises in the **northern foothills of Dandakaranya** near Sihawa in Raipur district of Chhattisgarh and runs through Orissa to discharge its water into the Bay of Bengal.



- 2 It is 851 km long and its catchment area spreads over 1.42 lakh sq. km.
- 3 Near the city of Sambalpur, it is dammed by the one of the largest earthen dam in the world, the **Hirakud Dam**.
- 4 Fifty three per cent of the drainage basin of this river lies in Madhya Pradesh and Chhattisgarh, while 47 per cent lies in Orissa.

MAHANADI RIVER

MADHYA PRADESH

JHARKHAND

RIGHT BANK TRIBUTARY

- Ong
- Tel
- Jonk

LEFT BANK TRIBUTARY

- Seonath
- Hasdeo
- Mand
- Ib

Gangrel Dam

Dudhawa Dam

Dandakaranya Hills
Chhattisgarh

Minimata (Hasdeo)
Bango Dam

Hirakud Dam

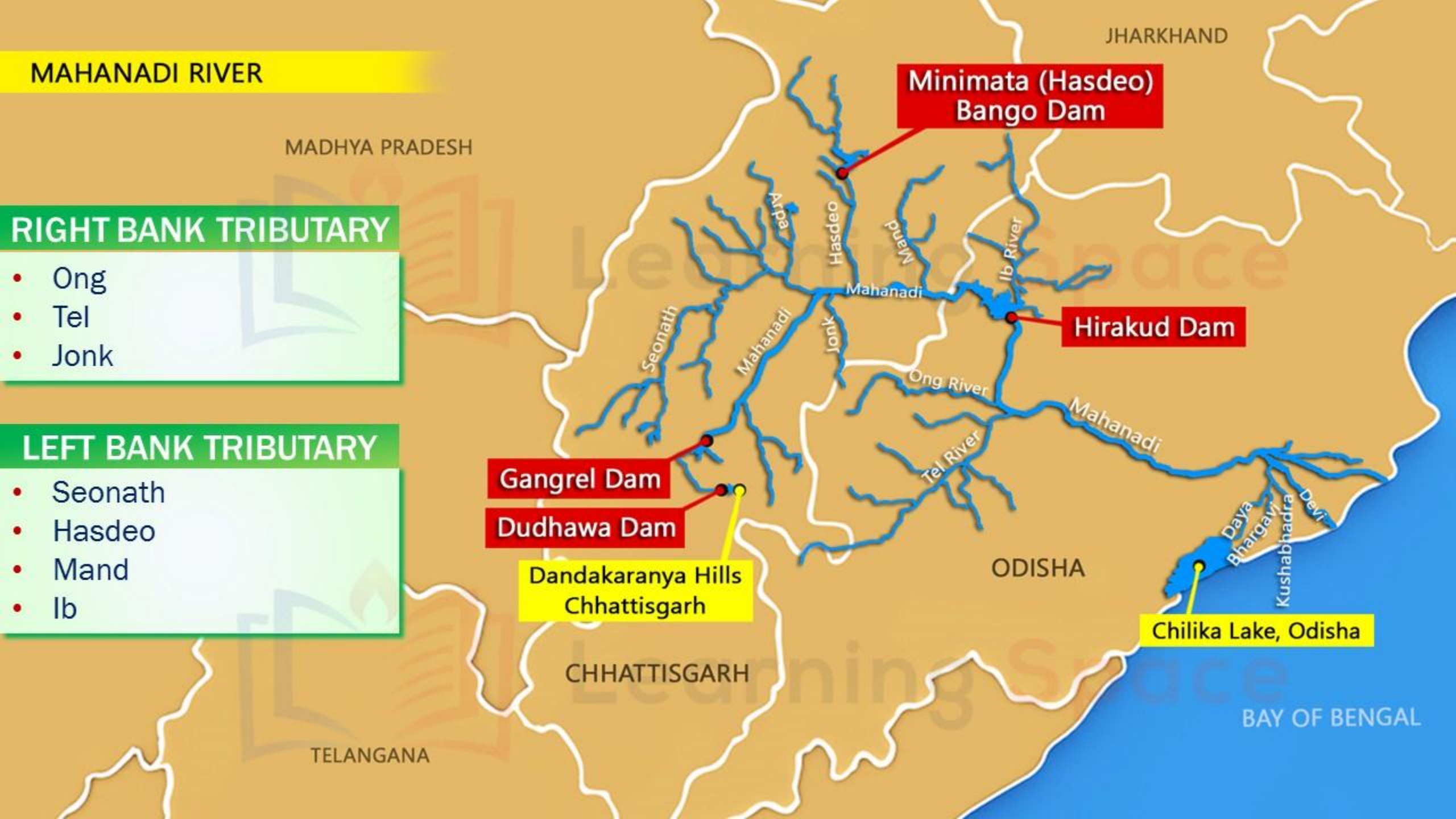
ODISHA

CHHATTISGARH

TELANGANA

Chilika Lake, Odisha

BAY OF BENGAL



THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

POINT NO. 2

THE GODAVARI

1 The Godavari is the **largest Peninsular river system**.

2 It is next only to the Ganga and the Indus system regarding sanctity, picturesqueness and is held in reverence as **Vridha Ganga or Dakshin Ganga**.

3 It rises in the **Triambakeshwar** Plateau of North Sahayadri hills near **Nasik district** of Maharashtra and discharges its water into the Bay of Bengal.

4 It is 1,465 km long with a catchment area spreading over 3.13 lakh sq.km. 49 per cent of this lies in Maharashtra and the rest lies in the states of

- Madhya Pradesh,
- Chattisgarh,
- Andhra Pradesh,
- Telangana,
- Orissa and
- Karnataka.

5 It receives a large number of tributaries from the left as well as from the right.

6 But the **left bank tributaries are more in number and larger in size** than the right bank tributaries

THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

POINT NO. 2

THE GODAVARI



- 7 The **Manjra** is the only important right bank tributary.

- 8 The **Penganga**, the **Wardha**, the **Wainganga**, the **Indravati**, the **Pranhita**, and the **Sabari** are its principal left bank tributaries.

- 9 The Godavari is subjected to heavy floods in its lower reaches to the south of Polavaram, where it forms a picturesque gorge.

- 10 It is navigable only in the deltaic stretch. The river after **Rajamundri** splits into several branches forming a large delta.



GODAVARI RIVER

THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

POINT NO. 3

THE KRISHNA

- 1 The Krishna is the second largest east flowing Peninsular river.
- 2 It rises near **Mahabaleshwar** in Sahyadri. Its total length is 1,401km.
- 3 The **Koyna**, the **Ghatprabha**, the **Malprabha**, the **Tungabhadra** and the **Bhima**, the **Musi** and the **Muneru** are its major tributaries.
- 4 The **Koyna** is a small tributary but is very famous for its **Koyna dam**. This dam was perhaps the main cause for the devastating **earthquake in 1967**.
- 5 At Wazirabad, it receives its last important tributary, the **Musi**, on whose banks the famous city of **Hyderabad** is located.
- 6 The total catchment area of Krishna river is spread across the states of Maharashtra, Karnataka, Andhra Pradesh and Telangana.
- 7 The Krishna delta appears to merge with that formed by the Godavari and extends about 35 km into the sea.

KRISHNA RIVER

CHHATTISGARH

ODISHA

MAHARASHTRA

TELANGANA

ANDHRA PRADESH

Near Jor Village,
Maharashtra

Ujjani Dam

Almatti Dam

Jurala Reservoir

Musi Reservoir

Pulichintala Dam

Dr. K. L. Rao Sagar
Pulichintala Reservoir

Prakasam Barrage

Hamsaladeevi

Narayanpur Dam

Nagarjuna Sagar

Srisaillam Dam

Sunkesula Barrage

Tungabhadra Reservoir

Vani Vilasa Sagar Dam

Bhadra Dam

KARNATAKA

ARABIAN SEA

BAY OF BENGAL



THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

POINT NO. 4

THE KAVERI

- 1 The source of the river is **Tala Cauvery** on the **Brahmagiri hills** (1,341m) of **Coorg district** in Karnataka.
- 2 Its length is 800 km and it drains an area of 81,155 sq. km.
- 3 This river is unique in the sense that its **upper catchment area** receives rainfall during **summer** by the south-west monsoon and the **lower catchment areas** during the **winter** season by the retreating north-east monsoon. It is therefore an “**almost**” **perennial river** with comparatively less fluctuations.
- 4 The river descends from the South Karnataka plains to the Tamilnadu plains through the **Sivasamudram** waterfalls.
- 5 The river divides itself into two distinct channels at **Srirangam**.
 - The northern channel is called **Kollidam**
 - While the southern channel retains the name **Cauvery**.
- 6 The delta formation starts from a point about 16 km away from Tiruchirappali.
- 7 Its basin area is spread across Karnataka, Kerala, Tamilnadu and Pondicherry.



NATIONAL PARKS - RIVER

Arabian Sea

Bay of Bengal

ANDHRA PRADESH

KARNATAKA

TAMIL NADU

Mudumalai National Park

▮ MOYAR RIVER, TRIBUTARY OF BHAVANI RIVER

Mukurthi National Park

➔ BHAVANI RIVER, TRIBUTARY OF CAUVERY RIVER

KERALA



NATIONAL PARKS - RIVER



THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

POINT NO. 5

THE NARMADA

- 1 The Narmada is the largest of all the west flowing rivers of the Peninsular India.
- 2 It originates on the western flank of the **Amarkantak plateau** at a height of about 1,057 m in the **Shahdol district** of Madhya Pradesh.
- 3 It flows through a **rift valley** between the **Satpura** in the south and the **Vindhyan** range in the north.
- 4 It forms a picturesque gorge which is popularly known as **Marble rocks** and here the world famous **Dhuan Dhar (Cloud of Mist)** waterfalls is formed near **Jabalpur**.
- 5 After flowing a distance of about 1,312 km, it meets the Arabian sea **south of Bharuch**, forming a broad 27 km long estuary before entering the **Gulf of Khambhat**.
- 6 Its catchment area is about 98,796 sq. km. The **Sardar Sarovar Project** has been constructed on this river.

NARMADA RIVER



RAJASTHAN

UTTAR PRADESH

MADHYA PRADESH

GUJARAT

Orsang

Heran

Indira Sagar Dam

Kolar

Rani Avanti Bai Sagar / Bargi Dam

Bharuch

Sardar Sarovar Dam

Omkareshwar Dam

Ganjal

Tawa

Tawa Reservoir

Shakkar

Amarkantak

MAHARASHTRA

CHHATTISGARH

THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

POINT NO. 6

THE TAPI



- 1 The Tapi is the second largest westward flowing river and is known as the **twin of the Narmada**.
- 2 It **originates from Multai** on the **Satpura plateau** in the **Betul** district of Madhya Pradesh.
- 3 It is 724 km long and drains an area of 65,145 sq. km.
- 4 Nearly 79 per cent of its basin lies in **Maharashtra**, 15 per cent in **Madhya Pradesh** and the remaining 6 per cent in **Gujarat**.



THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

POINT NO. 7

THE LUNI



1 Luni is the largest river system of Rajasthan, west of Aravali. It is also called as **Salt river or Lonari or Lavanavari**.

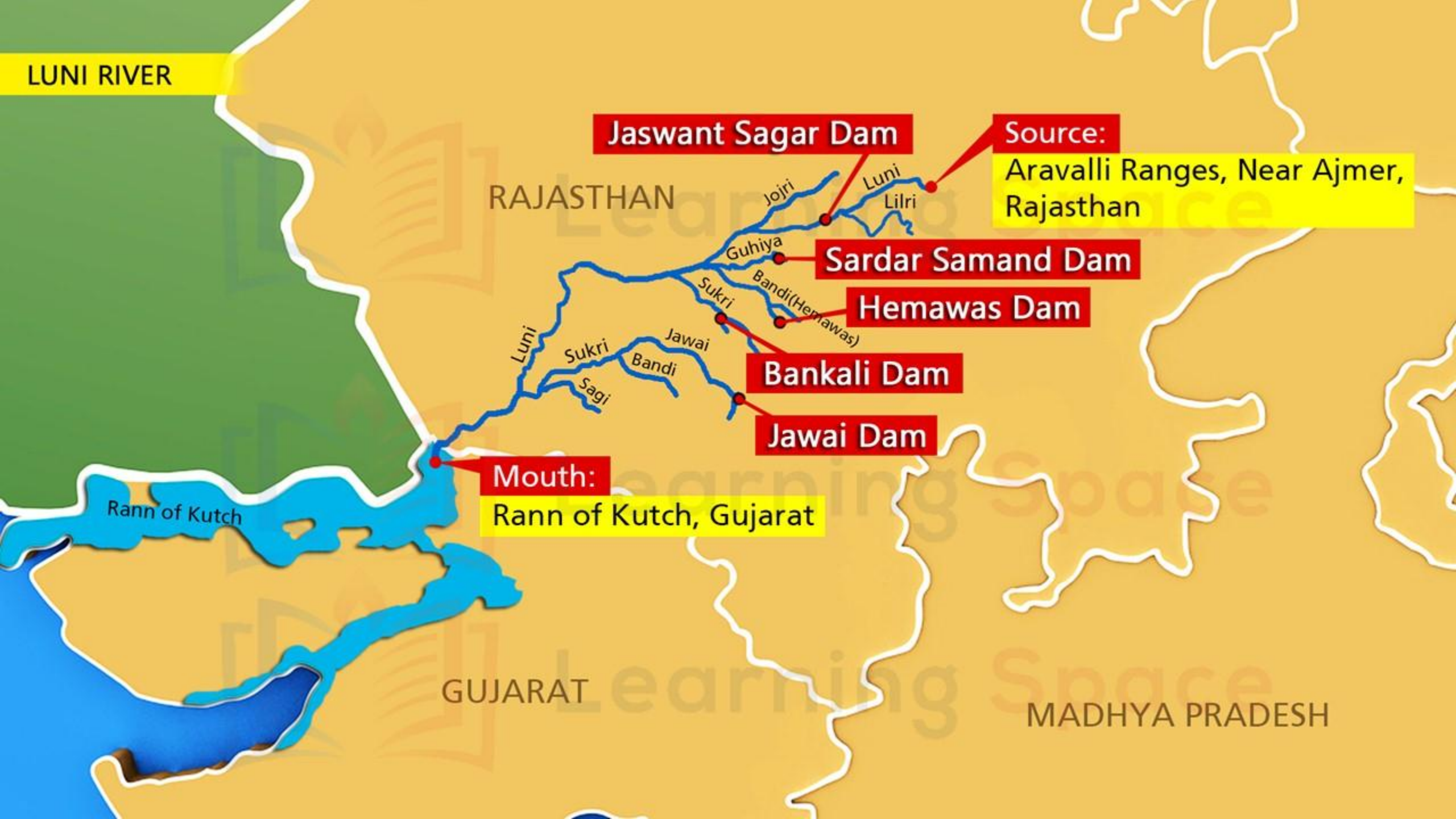
2 It is known for its **brackish** nature of water.

3 It originates near **Pushkar** in two branches which join with each other at Govindgarh.

4 From here, the river comes out of Aravali and is known as Luni.

5 It flows towards the west till Telwara and then takes a southwest direction and it is finally **lost in the marshy grounds at the head of Rann of Kuchchh**.

LUNI RIVER



INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

SMALL RIVERS FLOWING TOWARDS THE WEST

The rivers flowing towards the Arabian sea have short courses.

- 1 The **Shetrunji** is one such river which rises in Amreli district of Gujarat.
- 2 The **Bhadar** originates in Rajkot district .
- 3 **Sabarmati** and **Mahi** are the two famous rivers of Gujarat.
- 4 The **Vaitarna** rises from the Trimbak hills in Nasik district at an elevation of 670 m.
- 5 The **Kalinadi** rises from Belgaum district and falls in the Karwar Bay.
- 6 The source of **Bedti** river lies in Hubli Dharwar and traverses a course of 161km.
- 7 The **Sharavati** is another important river in Karnataka flowing towards the west.
- 8 The Sharavati originates in Shimoga district of Karnataka and drains a catchment area of 2,209 sq. km.

THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

SMALL RIVERS FLOWING TOWARDS THE WEST

10 The **Sabarmathi** is the name given to the combined streams of Sabar and Hathmati. It finally drains into the Gulf of Khambhat.

11 Goa has two important rivers which can be mentioned here. One is **Mandovi** and the other is **Juari**.



12 Kerala has a narrow coastline. The longest river of Kerala, **Bharathapuzha** rises near Annamalai hills. It is also known as **Ponnani**. It drains an area of 5,397 sq. km.

13 The **Periyar** is the second largest river of Kerala. Its catchment area is 5,243 sq. km.

14 Another river of Kerala worth mentioning is the **Pamba** river which falls in the **Vembanad lake** after traversing a course of 177 km.

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

SMALL RIVERS FLOWING TOWARDS THE EAST

1 There are small rivers which join the Bay of Bengal, though small, these are important in their own right.

2 The north-south arrangement of these rivers are Subarnrekha, the Baitarni, the Brahmani, the Vamsadhara, the Penneru, the Palar and the Vaigai are important rivers.

3 SUBARNAREKHA:

- The Subarnarekha originates from the Ranchi Plateau in Jharkhand at an elevation of 600m and flows in South-East direction forming the boundary between West Bengal and Orissa in its lower course.
- It joins Bay of Bengal forming an estuary between Ganga and Mahanadi deltas. Its total length is 395 kms.

THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

SMALL RIVERS FLOWING TOWARDS THE EAST

4 BRAHMANI RIVER:

- The Brahmani river comes into existence by the confluence of the **South Koel and Sankh rivers near Rourkela**. It has a total length of 800 kms.
- Its mouth becomes the northern end of the Mahanadi delta.

5 PENNURU RIVER:

- The Penneru river springs from the **Nandi Durg peak** in Karnataka and flows in northward direction.
- It then enters Andhra Pradesh and takes an easterly course.
- It forms a narrow estuary before it joins the Bay of Bengal. The total length of the river is 597 kms.

THE RIVER SYSTEMS OF THE PENINSULAR DRAINAGE

SMALL RIVERS FLOWING TOWARDS THE EAST



6

PONNAIYAR RIVER:

- The Ponnaiyar is a small stream which is confined to the coastal area only.

7

VAIGAI RIVER:

- South of the Cauvery delta, there are several streams, of which the Vaigai is the longest.
- This flows through dry channels and tends to get lost intermittently and appear again on the surface.

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